YouTube Content Success Analytics

From Trending Data to a Streamlit Strategy Dashboard

Pre-Joining Capstone Project | July 2026

Agenda

What we'll cover

1. Objective & Scope
2. Dataset Overview
3. Tool Stack
4. Data Quality & Cleaning
5. Key Engagement Drivers
6. Statistical Confidence
7. Demonstration
8. Insights & Recommendations

Problem Statement

Creators are guessing, not deciding

Millions of videos are uploaded every day, but almost no creator or brand can answer with evidence: **what actually makes a video trend, hold attention, and grow a channel?**

No shared benchmark for "success"

Signals scattered across raw, noisy logs

Content decisions made on gut feel, not evidence

Result: wasted uploads, inconsistent growth, and strategy that can't be defended with data.

Objectives

Why we did this analysis ?

To identify what actually drives **trending performance**, **audience retention** and **subscriber growth** and turn it into concrete content strategy moves.

Video-level engagement → **Content &** format levers → **Actionable** recommendations

Solution Overview

End-To-End High Level Approach

Raw Kaggle trending data → **Clean & deduplicate** → **Statistical Tests, EDA** → **Streamlit dashboard** → **Strategy recommendations**

One pipeline, one dataset, one dashboard from raw Kaggle export to a set of tested, explainable content-strategy levers.

Tool Stack

What we built the project with

Data Source	Environment	Analysis Libraries	Dashboard
Kaggle YouTube Trending Dataset	Google Colab (Python notebooks)	Pandas, Numpy	Streamlit (interactive app)
		scipy.stats (Statistical Tests)	
		Matplotlib	

Why this stack: Colab gave the team a shared, zero-setup workspace; Pandas/NumPy handled cleaning and stats at scale; Matplotlib and Seaborn produced the exploratory visuals; Streamlit turned the findings into a dashboard non-technical stakeholders can explore themselves.

Workflow / System Architecture

From raw file to end-user insight

Kaggle CSV (raw) → Colab note-book: clean, dedupe, stats → Processed dataset (36,540 videos) → Streamlit app: filters + charts → Creators / brand teams

Each stage is a separate, reproducible artefact: a versioned notebook feeds a clean CSV, which the Streamlit app reads directly — no manual copy-paste between steps.

Dataset Overview

The raw material

- **Source:** Kaggle — YouTube Trending Videos
- **Raw grain:** one row per (video, country, trending-day)
- **Coverage:** 100+ trending countries, 14 categories
- **Signals:** views, likes, comments, duration, definition, channel size, retention %

450K
raw trending rows

28
original columns

Data Schema at a Glance

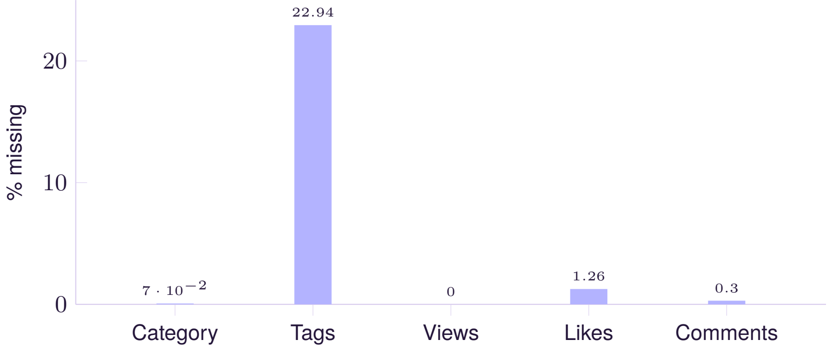
18 raw fields, grouped by role

Video Identity	Trending Signal	Engagement	Channel & Retention
video_id	video_trending_date	video_view_count	channel_country
video_published_at	video_trending_country	video_like_count	channel_view/sub_count
video_category_id		video_comment_count	channel_video_count
video_duration			avg_view_duration_sec
video_definition			avg_view_percentage

Cleaning decisions: Dropped `video_licensed_content` and low-signal free-text columns. Retained `channel_subscriber_count` (later grouped into size tiers). Retention metrics (`avg_view_duration_sec` and `avg_view_percentage`) are reserved for future watch-time analysis.

Data Quality — Completeness

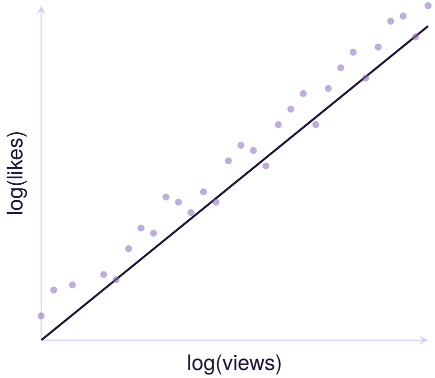
Missing values were real but small



Rows with missing engagement values or impossible records (likes > views) were removed.

Engagement Is Genuine

Likes and views move together



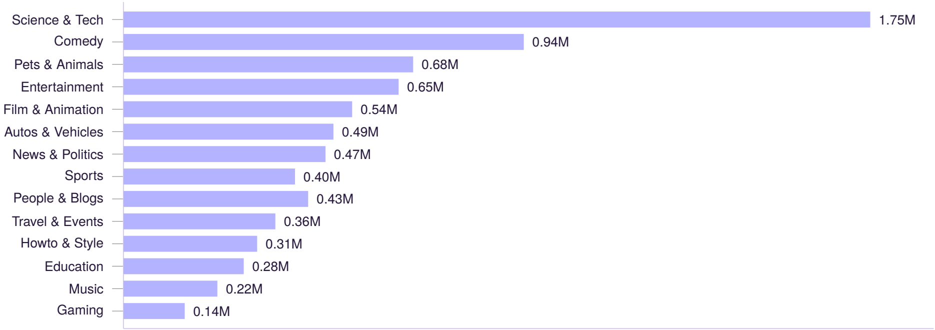
r = 0.87

correlation, log-log scale
($p \approx 0$)

- Confirms trending performance reflects **real audience response**, not artefacts
- Justifies using like-rate & engagement-rate as reliable success metrics

Performance by Category

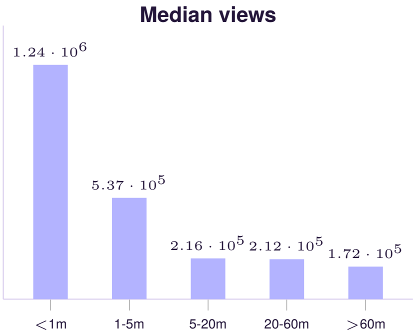
Median views per unique video

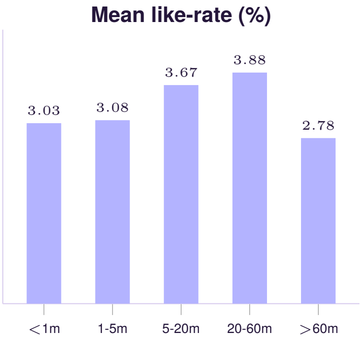


Science & Technology, Comedy and Pets & Animals over-index; Gaming and Music trend widely but reach far fewer viewers per video.

Reach vs. Engagement — Video Length

The barbell trade-off

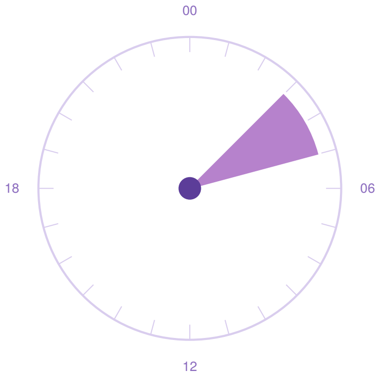




Shorts ($<$1 min) win **reach** ($\sim$6$\times$ the 5-20 min band); 20-60 min videos win **engagement**. Run both.

When to Publish

Hour of day shapes discovery



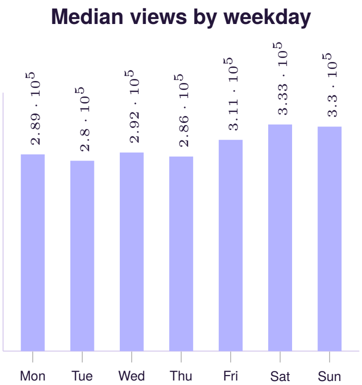
03:00 – 05:00

UTC: highest median trending views

- Aligns with waking hours across multiple large time zones simultaneously (8:30 pm to 10:30 pm IST)
- Like-rate varies only modestly by hour of day — timing mainly moves *reach*, not engagement quality

Publish Day Matters Too

Weekends win reach, weekdays hold engagement



- Weekend uploads (Sat/Sun) get the highest median views — roughly 15% above midweek
- But Saturday's mean like-rate (3.06%) is the week's *lowest*; weekdays hold steadier engagement (∼3.4–3.5%)
- Both effects are statistically significant (ANOVA, $p < 0.001$)

Use weekends to maximise reach; anchor "must-engage" content (launches, community asks) midweek.

Titles, Tags, Quality & Channel Size

Four smaller but real levers

Shorter titles win

Like-rate correlation: $\mathbf{r = -0.22}$

HD beats SD, decisively

Like-rate
$\mathbf{3.41\%}$ vs $\mathbf{1.63\%}$
$(p \approx 4 \times 10^{-10})$

More tags doesn't help

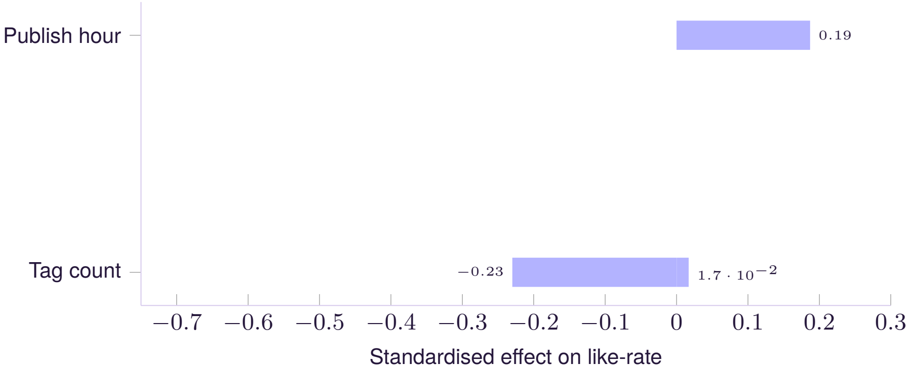
Tag count is slightly negatively correlated with views
$\mathbf{r = -0.08}$

Bigger channels reach further...

...but do not increase like-rate.
Subscriber count trends mildly negative once other factors are controlled.

What Really Drives Like-Rate

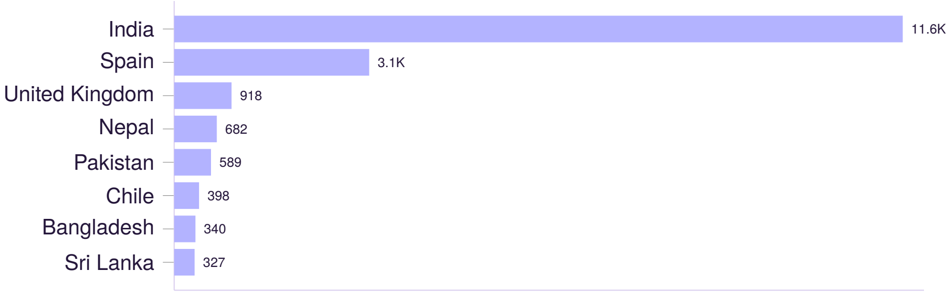
Multivariate view: each lever's effect net of the others



Title length is by far the strongest single lever (shorter is better); slow-to-trend and larger-channel videos under-perform on like-rate net of other factors. $R^2 \approx 0.07$ overall — real, but partial.

Geography: Reach Varies by an Order of Magnitude

Same platform, very different audiences by market



Median trending views by country (appearance grain). Nepal and Pakistan post far higher median views than larger markets like India or the UK — fewer videos compete for trending slots there, concentrating attention. Don't apply one global benchmark; localise success thresholds by market.

Category Mix Varies Sharply by Country

Where each content type actually trends

India dominates volume. Entertainment (5,459), Music (1,555) and Gaming (1,204) trend far more in India than any other market, a single country can skew global averages.

Sports is UK/Sri Lanka-heavy. 741 and 387 trending Sports videos respectively, far above their share of other categories.

Gaming spikes in Chile (340) & Bangladesh (198), disproportionate to their overall trending volume, a strong regional niche.

Statistical Confidence

The patterns are not noise

Views vs Likes

$\mathbf{r = 0.87}$

$p \approx 0$

HD vs SD

3.41% vs 1.63%

$p \approx 4 \times 10^{-10}$

Category / Format / Weekday

All differ

Very small p-values

$R^2 \approx 0.07$

of like-rate variance explained by content metadata alone

Business Recommendations (1/3)

Reach & format strategy

1. **Lead with Shorts for reach.** Sub-1-min videos hit $\sim$1.24M median views — $\sim$6$\times$ the 5-20 min band.

2. **Use mid-to-long videos to deepen engagement.** Like-rate peaks at 20-60 min ($\sim$3.9%) vs Shorts ($\sim$3.0%). Run a barbell strategy.

3. **Prioritise high-value categories.** Science & Tech, Comedy, Pets & Animals over-index; Gaming, Music, Education trail per video.

Business Recommendations (2/3)

Execution details

4. Publish in the early-UTC window. Uploads around 03:00-05:00 UTC show the highest median trending views.

5. Always ship HD HD like-rate (3.4%) far exceeds SD (1.6%).

6. Don't over-tag. More tags do not increase views. A focused, relevant tag set is sufficient.

Business Recommendations (3/3)

Deeper, multivariate-confirmed levers

7.Don't run one global content-category strategy. Category mix varies by an order of magnitude across markets. India over-indexes on Entertainment/Music/Blogs, UK and Sri Lanka on Sports, Chile and Bangladesh on Gaming.

8. Split weekday/weekend roles. Publish reach-plays on weekends (highest median views); save always-engage content (launches, asks) for weekdays (steadier like-rate).

9. Localise success benchmarks by market. Median views differ by $10\times+$ across countries (e.g. Nepal/Pakistan vs India/UK). Don't judge every market against one global number.

Challenges & Learnings

Being honest about the rough edges

Data challenge

~23% of rows had missing tags, and some rows showed likes $>$ views, both were cleaned before analysis.

Analytical challenge

One video trended 1,558 times across 81 countries; deduplicating without losing genuine reach took iteration.

Key learning

Metadata alone explains only $R^2 \approx 0.07$ of like-rate: real, but partial. Topic and creator identity matter just as much.

Takeaway: trending-level data narrows the odds of success, it does not guarantee a hit, and results should be validated against each channel's own analytics.

Conclusion & Future Scope

From findings to a 90-day pilot

1

Pilot

Shorts + long-form barbell on 2-3 channels

2

Test

A/B the 03:00–05:00 UTC window & weekday/weekend split

3

Extend

Retention/watch-time study + GenAI Q&A layer on the dashboard

Read this with one caveat in mind: findings are based on *trending* videos only ($R^2 \approx 0.07$ for content metadata). validate against each channel's own analytics before committing budget.

Thank You

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